

Shafar Foundation presents

Battle of Brains 2026

Problemset



Date: August 17, 2026

A. The Forbidden Symphony

Time Limit: 1s

Memory Limit: 256MB

The royal chief composer wants to compose a grand symphony. This symphony will be a string of exact length L , consisting only of lowercase English letters (a-z).

However, by the order of the king, exactly N specific “Forbidden Melodies” have been banned. This means that none of these N substrings can appear anywhere in the symphony composed by the composer.

You need to find out how many different valid symphonies can be composed following these rules. Since this number can be very large, print the answer modulo $10^9 + 7$.

Input

The first line contains two integers N and L ($1 \leq N \leq 100$, $1 \leq L \leq 10^{18}$) — the number of forbidden melodies and the exact length of the symphony, respectively.

Each of the next N lines contains a string of lowercase English letters, representing a forbidden melody.

The total length of all forbidden melodies (sum of lengths of all strings) will be at most 100.

Output

Print a single integer representing the number of valid symphonies modulo $10^9 + 7$.

Example

Input	Output
2 3 ab ba	17474

B. Stock Market Crash

Time Limit: 1s

Memory Limit: 256MB

The stock market has crashed. Two rival investors, Alice and Bob, are playing a game. There are N companies' shares in the stock market, and their current prices are $A_1, A_2, \dots, A_N$ respectively.

The rules of the game are:

1. Alice and Bob take turns to make a move. Alice always moves first.
2. In each move, an investor must choose a company's share whose price is $A_i > 0$.
3. Then, the investor chooses a positive integer X and updates the price of the chosen share to $A_i \& X$ (Bitwise AND).
4. The condition is that the new price must be strictly less than the previous price. (i.e., $A_i \& X < A_i$).
5. The investor who cannot make a move (i.e., when all share prices become 0) goes bankrupt and loses the game.

If both investors play optimally, who will win?

Input

The first line contains an integer T ($1 \leq T \leq 10^4$) — the number of test cases.

For each test case:

- The first line contains an integer N ($1 \leq N \leq 10^5$).
- The second line contains N space-separated integers $A_1, A_2, \dots, A_N$ ($0 \leq A_i \leq 10^{18}$).

It is guaranteed that the sum of N across all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, print the name of the winning investor: "Alice" or "Bob".

Example

Input	Output
3	Alice
2	Bob
3 4	Bob
2	
3 5	
3	
0 0 0	

Note

In the first test case, $A = \{3, 4\}$. The binary representation of 3 is 011 and 4 is 100. If Alice plays optimally, she will win.

In the second test case, $A = \{3, 5\}$. The binary representation of 3 is 011 and 5 is 101. Here Bob will win.

In the third test case, all share prices are already 0 from the beginning. So Alice cannot make the first move, and Bob wins immediately.

C. Hello, Contestants!

Time Limit: 1s

Memory Limit: 256MB

Welcome to the programming contest! Your task is very simple. You just need to print the following message exactly as shown:

```
Welcome to Battle of Brains 2026!  
Good luck, contestants!
```

Input

There is no input.

Output

Print exactly:

```
Welcome to Battle of Brains 2026!  
Good luck, contestants!
```

Example

Input	Output
	Welcome to Battle of Brains 2026! Good luck, contestants!

D. Rafat's Business (Easy Version)

Time Limit: 1s

Memory Limit: 256MB

Rafat has a corporate job, but due to the extreme work pressure and his boss's strictness, he wants to quit. Seeking advice, he talks to his friend Naim, who suggests starting a business.

Rafat rents a new shop and decorates it. However, because his shop is quite small, he can stock a maximum of exactly k items in it.

To buy goods for his business, Rafat goes to a wholesaler. The wholesaler provides him with a list of available items, their quantities, and their unit prices. The wholesaler has a strict rule: if Rafat decides to buy a particular type of item, he must buy **all** the stock the wholesaler has for that item.

Rafat wants to fill exactly his shop's capacity (exactly k items). What is the minimum total cost required to completely fill his shop's stock?

Input

The first line contains an integer t ($1 \leq t \leq 10^4$) — the number of test cases.

For each test case:

- The first line contains two integers n and k ($1 \leq n, k \leq 10^5$) — the total number of item types available at the wholesaler, and the exact capacity of Rafat's shop.
- The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 2$) — where a_i represents the quantity (stock) of the i -th item available at the wholesaler.
- The third line contains n integers $b_1, b_2, \dots, b_n$ ($1 \leq b_i \leq 10^9$) — where b_i is the unit price of the i -th item. This means buying the whole stock of the i -th item will cost $a_i \times b_i$.

It is guaranteed that the sum of n across all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, print a single integer — the minimum amount of money Rafat needs to spend to buy exactly k items.

If it is not possible to buy exactly k items, print -1 .

Example

Input	Output
3	40
3 3	-1
1 2 1	60
10 15 20	
2 3	
2 2	
5 5	
4 4	
2 1 2 1	
10 100 20 200	

Note

In the first test case, the items are:

1. Quantity 1, Unit Price 10, Total Cost = 10
2. Quantity 2, Unit Price 15, Total Cost = 30
3. Quantity 1, Unit Price 20, Total Cost = 20

To get exactly 3 items, Rafat can pick item 1 and item 2. Total items = $1 + 2 = 3$. Total cost = $10 + 30 = 40$. (Picking item 2 and item 3 would cost $30 + 20 = 50$, which is higher).

In the second test case, Rafat needs exactly 3 items, but both available items have a quantity of 2. It is impossible to get exactly 3 items. Hence, the output is -1.

E. Rafat's Business (Hard Version)

Time Limit: 1s

Memory Limit: 256MB

Rafat has a corporate job, but due to the extreme work pressure and his boss's strictness, he wants to quit. Seeking advice, he talks to his friend Naim, who suggests starting a business.

Rafat rents a new shop and decorates it. However, because his shop has a limited space, he can stock exactly k items in it.

To buy goods for his business, Rafat goes to a wholesaler. The wholesaler provides him with a list of available items, their quantities, and their unit prices. The wholesaler has a strict rule: if Rafat decides to buy a particular type of item, he must buy **all** the stock the wholesaler has for that item.

Rafat wants to fill exactly his shop's capacity (exactly k items). What is the minimum total cost required to completely fill his shop's stock?

Input

The first line contains an integer t ($1 \leq t \leq 100$) — the number of test cases.

For each test case:

- The first line contains two integers n and k ($1 \leq n \leq 1000$, $1 \leq k \leq 10^4$) — the total number of item types available at the wholesaler, and the exact capacity of Rafat's shop.
- The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 100$) — where a_i represents the quantity (stock) of the i -th item available at the wholesaler.
- The third line contains n integers $b_1, b_2, \dots, b_n$ ($1 \leq b_i \leq 10^9$) — where b_i is the unit price of the i -th item. This means buying the whole stock of the i -th item will cost $a_i \times b_i$.

It is guaranteed that the sum of n across all test cases does not exceed 5000.

Output

For each test case, print a single integer — the minimum amount of money Rafat needs to spend to buy exactly k items.

If it is not possible to buy exactly k items, print -1.

Example

Input	Output
3	65
3 5	-1
2 3 5	200
10 15 20	
2 3	
2 2	
5 5	
4 7	
2 1 4 3	
10 100 20 200	

Note

In the first test case, the items are:

1. Quantity 2, Unit Price 10, Total Cost = 20
2. Quantity 3, Unit Price 15, Total Cost = 45
3. Quantity 5, Unit Price 20, Total Cost = 100

To get exactly 5 items, Rafat can pick item 1 and item 2 ($2 + 3 = 5$). Total cost = $20 + 45 = 65$. Or he can pick item 3. Total cost = 100. The minimum cost is 65.

In the second test case, Rafat needs exactly 3 items, but both available items have a quantity of 2. It is impossible to get exactly 3 items. Hence, the output is -1.

F. All in one Gun

Time Limit: 2s

Memory Limit: 256MB

You are developing a new shooter game, but since there are many shooter games already, you decide to make your weapon unique.

You have an all-in-one gun with n bullets in its magazine. The i -th bullet deals a_i damage. The enemy starts with h health and dies as soon as its health becomes less than or equal to 0.

The gun fires exactly one bullet per second. After all n bullets have been fired, the gun must reload, which takes k seconds. Reloading always restores exactly the same sequence of bullets. You cannot reload early. At the beginning of the fight, the magazine is already full.

Before the fight starts, you may perform at most one swap: choose two different indices i and j and exchange a_i and a_j . You may also choose not to swap anything.

Your task is to find the minimum number of seconds needed to kill the enemy, assuming you choose the best possible optional swap.

Input

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

For each test case:

- The first line contains three integers n , h , and k ($2 \leq n \leq 2 \cdot 10^5$, $1 \leq h \leq 10^{18}$, $1 \leq k \leq 10^9$) — the number of bullets, the enemy's initial health, and the reload time in seconds.
- The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^9$), where a_i is the damage dealt by the i -th bullet.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, print one integer — the minimum number of seconds required to kill the enemy.

Example

Input	Output
6	3
5 10 1	2
4 2 3 5 3	7
5 10 1	6
4 2 3 7 3	19
3 10 2	17
1 2 3	
2 5 3	
2 1	
3 18 5	
1 2 3	
4 10 10	
1 1 2 2	

Note

In the first test case, the original magazine is $[4, 2, 3, 5, 3]$. Swap the bullets at positions 2 and 5 to obtain $[4, 3, 3, 5, 2]$. The first three bullets deal $4 + 3 + 3 = 10$ damage, so the enemy dies after 3 seconds.

In the third test case, swapping the first and third bullets changes $[1, 2, 3]$ to $[3, 2, 1]$. The first magazine deals 6 damage. After 2 seconds of reloading, the first two bullets of the next magazine deal another 5 damage. Thus the enemy dies after $3 + 2 + 2 = 7$ seconds.

G. Elite Network Routing

Time Limit: 1s

Memory Limit: 256MB

There are N servers in a massive corporate network. These servers are connected to each other by M bidirectional data cables. Each cable has a specific “Vulnerability Score” or risk level of being hacked, which is represented by an integer.

You need to send a highly confidential data packet from server 1 to server N . When sending data through a specific path, the “risk” of that path is the **maximum** Vulnerability Score among all the cables present in that path.

Since the data is highly important, you need to find a path such that the maximum risk of the path is as **low** as possible. If it is impossible to reach server N from server 1 by any path, print -1.

Input

The first line contains two integers N and M ($2 \leq N \leq 10^5$, $1 \leq M \leq 2 \cdot 10^5$) — the number of servers and the number of cables, respectively.

Each of the next M lines contains three integers u, v, w ($1 \leq u, v \leq N$, $1 \leq w \leq 10^9$, $u \neq v$), indicating that there is a cable between server u and server v with a Vulnerability Score of w .

Output

Print a single integer representing the minimum possible maximum risk among all paths from server 1 to server N . If no path exists, print -1.

Example

Input	Output
<pre> 5 6 1 2 10 1 3 20 2 4 50 3 4 15 4 5 30 3 5 40 </pre>	<pre> 30 </pre>

Note

There are multiple paths to go from server 1 to server 5:

- Path 1: $1 \rightarrow 2 \rightarrow 4 \rightarrow 5$. The scores of the edges in this path are 10, 50, and 30. The maximum risk in this path is 50.
- Path 2: $1 \rightarrow 3 \rightarrow 5$. The scores are 20 and 40. The maximum risk here is 40.
- Path 3: $1 \rightarrow 3 \rightarrow 4 \rightarrow 5$. The scores are 20, 15, and 30. The maximum risk here is 30.

Among all paths, the 3rd path has the lowest maximum risk (30). Therefore, the answer is 30.

H. The Final Decision

Time Limit: 1s

Memory Limit: 256MB

As part of the Battle of Brains 2026, the organizing committee decided to conduct an experimental activity to determine how well contestants can understand instructions.

For this activity, each contestant is given two positive integers, A and B . These numbers are not necessarily related to each other, and their positions in the input do not indicate any special meaning.

The committee has designed a special calculation procedure.

- First, the machine examines the relationship between the two numbers. If the two numbers happen to represent the same value, the machine considers them to belong to the same category. In this situation, instead of combining their values directly, the machine uses both values together in a multiplication operation.
- On the other hand, if the two numbers do not represent the same value, they are considered to belong to different categories. In that case, the machine combines their values using an addition operation.

The machine does not perform both operations. It performs exactly one of them according to the relationship between A and B .

Input

The single line of input contains two positive integers A and B ($1 \leq A, B \leq 1000$).

Output

Print a single integer representing the final value produced by the machine.

Examples

Input	Output
4 9	13
7 7	49

I. Triple Key Authentication

Time Limit: 1s

Memory Limit: 256MB

To access the main server of pcIST, a “three-factor authentication” system has been designed. To unlock this system, an admin must input three integers x, y, z .

According to the rules of the system:

1. Each number must be between 1 and N inclusive. That means, $1 \leq x, y, z \leq N$.
2. The greatest common divisor (GCD) of the three numbers must be exactly 1. That means, $\gcd(x, y, z) = 1$.

Note that the numbers do not necessarily have to be pairwise coprime. For example, $(6, 10, 15)$ is a valid triplet because although $\gcd(6, 10) = 2$, $\gcd(10, 15) = 5$, and $\gcd(6, 15) = 3$, their overall GCD is $\gcd(6, 10, 15) = 1$.

You need to find out how many different valid triplets (x, y, z) can be formed to unlock the system. Since this number can be very large, print the answer modulo $10^9 + 7$.

Input

The first line contains an integer T ($1 \leq T \leq 10^4$) — the number of test cases.

Each of the next T lines contains a single integer N ($1 \leq N \leq 10^6$).

Output

For each test case, print a single integer — the number of valid triplets, modulo $10^9 + 7$.

Example

Input	Output
2	7
2	181
6	

Note

In the first test case ($N = 2$), the valid triplets are:

$(1, 1, 1), (1, 1, 2), (1, 2, 1), (2, 1, 1), (1, 2, 2), (2, 1, 2), (2, 2, 1)$

There are exactly 7 valid triplets.

J. Festival Lights

Time Limit: 2s

Memory Limit: 512MB

Alice, Bob, and Carol are preparing lights for a festival. The festival lasts for m days, numbered from 1 to m .

Alice turns on her light every a days (on days $a, 2a, 3a, \dots$). Bob turns on his light every b days, and Carol turns on her light every c days.

Whenever exactly one person turns on their light on a day, that person receives 6 points. If exactly two people turn on their lights on the same day, the 6 points are divided equally, so each receives 3 points. If all three turn on their lights on the same day, each receives 2 points.

Your task is to calculate the total number of points collected by Alice, Bob, and Carol during the first m days.

Input

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

Each test case contains four integers a, b, c , and m ($1 \leq a, b, c \leq 10^6, 1 \leq m \leq 10^{17}$).

Output

For each test case, print three integers — the total number of points collected by Alice, Bob, and Carol, respectively.

Example

Input	Output
5	24 15 9
2 3 5 10	34 10 4
1 2 4 8	38 14 8
3 6 9 30	68 41 23
4 6 10 60	78 51 39
7 11 13 100	

Note

For the first test case, $a = 2$, $b = 3$, $c = 5$, and $m = 10$.

Alice visits on days 2, 4, 6, 8, 10; Bob visits on days 3, 6, 9; and Carol visits on days 5, 10.

- On day 6, Alice and Bob visit together, so each receives 3 points.
- On day 10, Alice and Carol visit together, so each receives 3 points.

All other visiting days in this range have only one visitor.

Therefore, the totals are 24, 15, and 9.